Caitlin Wilks

LEAD DEVELOPER

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Lead Developer with over a decade shipping high-performance software across 3D visualisation, geographic data systems, and cloud platforms. A genuine technology polyglot — equally at home in **Rust**, **C++**, **C#**, and **Python** — with a record of taking ambiguous, technically hard problems from first principles through to production. Currently leading a team of six at Virtalis, pairing hands-on engineering with technical direction, and using AI-assisted development — agentic coding tools and LLM-based code review — to deliver full end-to-end, shippable features and software.

SKILLS

Languages & Compute
Rust, C#, C++, Python, Haskell, CUDA / GPGPU

Cloud & Infrastructure
Kubernetes, Docker, Azure, RabbitMQ, CI/CD

Architecture & Leadership
Microservices, CRUD / CQRS, React, cross-platform; team leadership, mentoring, Agile/Jira

AI Integration & AI-Assisted Engineering
MCP servers, AI assistant integration, small language models, agentic coding tools, LLM code & security review

EXPERIENCE

Lead Developer *Virtalis Ltd* — Apr 2026 – Present

Returned to Virtalis to take on a direct leadership role, building the next generation of the company's high-performance 3D visualisation products for manufacturing and industrial clients — a boomerang appointment following an earlier eight-year run at the company from 2015 to 2022.

- Leads a team of six developers end-to-end — setting technical direction, unblocking the hardest problems personally, and owning delivery on key projects.
- Building AI assistant integration into the company's software, including a Model Context Protocol (MCP) server exposing product functionality to LLM-based assistants, and experimenting with small language models to deliver specific, targeted capabilities.
- Assesses new tools and techniques for feasibility, including where AI-assisted development can responsibly fit into the team's workflow, before recommending adoption.
- Uses agentic coding tools such as Claude Code to rapidly prototype new product concepts, and LLM-based analysis to review new code for structural weaknesses and security vulnerabilities.

Senior Software Engineer *Map Impact Ltd* — Jan 2024 – Mar 2026

Developed high-availability, high-performance cloud GIS software using Azure, Kubernetes, Rust, Python, C# and React — producing geographic datasets for habitat surveying used by local councils and land developers, and the software that delivered them to customers.

- Designed and built services to deliver large geographic datasets to customers — including a custom web feature service benchmarked against GeoServer, and a web-based GIS application for customers to view, operate on, and export their own data.
- Worked with in-house scientists to build and automate data-processing pipelines, keeping datasets current as new satellite data arrived.
- Mentored junior engineers on writing production-ready code and shipping it, and broke larger projects into manageable stories and tasks.

Senior Software Engineer *Virtalis Ltd* — Jun 2015 – Oct 2022

Built high-performance desktop, distributed, and cloud-based 3D visualisation software for manufacturing and geoscience customers over an eight-year run — moving from individual contributor to trusted owner of the toughest technical problems in the business.

- Directed the design and development of a large new cloud-based product to replace the company's conventional desktop offering, with an emphasis on high-performance backend services.
- Migrated the development team from an ageing, self-hosted SVN setup to modern Git-based development on Azure, including CI/CD, unit testing, and a GitFlow branching strategy.
- Built custom solutions for customers' 3D visualisation needs, including a real-time volumetric sky with clouds for flight-training simulators, and support for large, custom streaming data formats.
- Integrated third-party VR and input hardware into the visualisation suite, occasionally designing custom input devices for specialist use cases.
- Rendered geographic and geological data onto 3D terrains and other visualisations, working across C++, C#, Kubernetes, and RabbitMQ in desktop and microservice architectures.

EDUCATION

BSc, Computer Games Programming

Dissertation: *"Real-time GPU Ray Tracing with Heterogeneous Volumetric Effects"* — direct 3D visualisation of volumetric data on the GPU. Available at cat.bio/dissertation.pdf

BEYOND WORK

- **Better Tactics** — a chess tactics trainer written from scratch in Rust, using spaced repetition to build calculation skill and pattern recognition.
- **Flex** — a fast, fully local media player and organiser in Rust on libmpv, designed for TV-remote use.
- **Talkhaus** — runs a self-hosted Kubernetes cluster for a gaming community's forum and web services.
- **Pillowdrift** — co-founded a game studio with three friends during university, running it full-time for a year and shipping original games plus freelance work.
- Occasional technical partner on game translation and modding projects, reverse-engineering titles to add features or language support.